

Summary

This thesis examines fine-tuning small open-weight language models for binary email classification as legitimate or unwanted. After reviewing the available models, Qwen3-0.6B was selected for the experiments. Low-rank adaptation (LoRA) was used. Sixteen configurations and model states saved at successive training stages were compared. Four ways of using the model as a classifier were then evaluated. The comparison also included the unmodified model and six other classifiers. The final evaluation used four disjoint test sets containing 2,000 messages each. Across the four test sets, the best fine-tuned variant achieved a mean classification accuracy of 97.8%. DistilBERT, the most accurate of the other classifiers, achieved a mean accuracy of 94.4% and processed messages approximately 2.3 times faster than Qwen3. Using the fine-tuned model in an email filter requires accounting for the computational cost of classification and evaluating it on current email traffic.